

M T W T F S

DATE: \_\_\_\_\_

Ass 02

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Q:-

$$A = \begin{bmatrix} 1 & 5 & 2 \\ 1 & 1 & 7 \\ 0 & -3 & 4 \end{bmatrix} \quad x = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} \quad b = \begin{bmatrix} 9 \\ 6 \\ -2 \end{bmatrix}$$

$$A_{b.} = \begin{bmatrix} 1 & 5 & 2 & 9 \\ 1 & 1 & 7 & 6 \\ 0 & -3 & 4 & -2 \end{bmatrix}$$

$$R = \begin{bmatrix} 1 & 5 & 2 & 9 \\ 0 & -4 & 5 & -3 \\ 0 & -3 & 4 & -2 \end{bmatrix} \quad \text{by } R_2 - R_1$$

$$R = \begin{bmatrix} 1 & 5 & 2 & 9 \\ 0 & 1 & -5/4 & 3/4 \\ 0 & -3 & 4 & -2 \end{bmatrix} \quad \text{by } (-1/4) R_2$$

$$R \left[ \begin{array}{ccc|c} 1 & 5 & 2 & 9 \\ 0 & 1 & -5/4 & 3/4 \\ 0 & 0 & 1/4 & 1/4 \end{array} \right] \text{ by } R_3 + 3R_2$$

$$R \left[ \begin{array}{ccc|c} 1 & 5 & 2 & 9 \\ 0 & 1 & -5/4 & 3/4 \\ 0 & 0 & 1 & 1 \end{array} \right] \text{ by } 4R_3$$

$$x_1 + 5x_2 + 2x_3 = 9 \quad \text{--- (i)}$$

$$x_2 - 5/4 x_3 = 3/4 \quad \text{--- (ii)}$$

$$\boxed{x_3 = 1} \quad \text{--- (iii)}$$

Put  $x_3 = 1$  in eq. (ii)

Put  $x_3 = 1, x_2 = 2$  in eq. (i)

$$x_2 = \frac{5(1) + 3}{4} = \frac{8}{4}$$

$$x_1 + 5(2) + 2(1) = 9$$

$$x_1 + 12 = 9$$

$$x_2 = \frac{3 + 5}{4} = \frac{8}{4}$$

$$x_1 = 9 - 12$$

$$\boxed{x_1 = -3}$$

$$\boxed{x_2 = 2}$$

$$x_1 = -3$$

$$x_2 = 2$$

$$x_3 = 1$$